- $\forall k \in \mathbb{N}$

$$\begin{aligned}
\operatorname{Cov}(R_t, R_{t-k}) &= \operatorname{Cov}(\cos(ct) Z_t + \sin(ct) Z_{t-1}, \\
&\qquad \cos(c(t-k)) Z_{t-k} + \sin(c(t-k)) Z_{t-k-1}) \\
&= \cos(ct)\cos(c(t-k))\operatorname{Cov}(Z_t, Z_{t-k}) \\
&+ \cos(ct)\sin(c(t-k))\overbrace{\operatorname{Cov}(Z_t, Z_{t-k-1})}^{0} \\
&+ \sin(ct)\cos(c(t-k))\operatorname{Cov}(Z_{t-1}, Z_{t-k}) \\
&+ \sin(ct)\sin(c(t-k))\operatorname{Cov}(Z_{t-1}, Z_{t-k-1}) \\
&= \begin{cases} (\cos^2(ct) + \sin^2(ct))\,\sigma^2 = \sigma^2, & \text{if } k = 0 \\ \sin(ct)\cos(c(t-1))\,\sigma^2, & \text{if } k = 1 \\ 0, & \text{else} \end{cases}
\end{aligned}$$

$\gamma_R(k)$ is only a function of k and t

Conclusion: $(R_t)_t$ is not stationary